

Molecular Dynamics Simulations

Computational Physics II - Lecture 7

Giovanni Settanni and Marialore Sulpizi

June 9, 2026

Where We Are in the Course

Previous lectures – Monte Carlo

- ▶ Random numbers & probability distributions
- ▶ MC integration, importance sampling
- ▶ Markov processes & master equation
- ▶ Metropolis algorithm
- ▶ Ising model, critical slowing-down
- ▶ Wolff, parallel tempering, Wang-Landau

New module – Molecular Dynamics

- ✓ **Today:** MD fundamentals + LJ fluid
- **Next:** Thermostats & barostats

Key question

MC samples *configurations*.

What if we also want *dynamics*?

⇒ Follow Newton's equations.

MD vs. MC

MC: efficient equilibrium thermodynamics (no real time).

MD: time-resolved dynamics and transport properties.

What Is Molecular Dynamics?

Definition

Molecular Dynamics (MD) numerically integrates Newton's classical equations of motion for N interacting particles, generating time-resolved trajectories $\{\mathbf{r}_i(t), \mathbf{v}_i(t)\}$.

What we compute

- ▶ Trajectories $\mathbf{r}_i(t), \mathbf{v}_i(t)$
- ▶ Time avg. $\leftrightarrow$ ensemble avg. (via ergodic hypothesis)
- ▶ Dynamics: diffusion, viscosity
- ▶ Structure: $g(r), S(q)$

Typical applications

- ▶ Materials science
- ▶ Biomolecular systems
- ▶ Liquids & glasses
- ▶ Colloidal suspensions
- ▶ Active matter

Historical note

First MD: Alder & Wainwright (1957) – hard spheres.

First continuous potential: Rahman (1964) – liquid Argon (LJ).

The Ergodic Hypothesis

Statistical mechanics: observables are ensemble averages

$$\langle A \rangle_{\text{ens}} = \int A \rho(\mathbf{r}^N, \mathbf{p}^N) d\mathbf{r}^N d\mathbf{p}^N$$

Ergodic hypothesis: for a long trajectory

$$\langle A \rangle_{\text{time}} = \langle A \rangle_{\text{ens}}$$

$$\langle A \rangle_{\text{time}} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T A(\mathbf{r}^N(t), \mathbf{p}^N(t)) dt$$

Caveat

May fail for:

- ▶ Glassy / jammed systems
- ▶ Metastable states
- ▶ Too-short runs

Checks for Convergence always necessary.

Newton's Equations of Motion

For N particles with masses m_i and positions $\mathbf{r}_i$:

$$m_i \ddot{\mathbf{r}}_i = \mathbf{F}_i = -\nabla_i U(\mathbf{r}_1, \dots, \mathbf{r}_N)$$

Hamiltonian formulation

$$H = \sum_{i=1}^N \frac{\mathbf{p}_i^2}{2m_i} + U(\mathbf{r}^N) = E$$

$$\dot{\mathbf{r}}_i = \frac{\partial H}{\partial \mathbf{p}_i}, \quad \dot{\mathbf{p}}_i = -\frac{\partial H}{\partial \mathbf{r}_i}$$

$H = E = \text{const}$ along any trajectory.

Instantaneous temperature (via equipartition theorem):

$$\langle k_B T \rangle = \langle m_i v_{ci}^2 \rangle \implies T = \frac{1}{k_B N_{\text{dof}}} \sum_i \sum_c m_i v_{ci}^2 = \frac{1}{k_B 3N} \sum_i m_i v_i^2 = \frac{2K}{k_B 3N}$$

Conserved quantities (NVE)

- ▶ Total energy $E = K + U$
- ▶ Total momentum $\mathbf{P} = \sum_i \mathbf{p}_i$
- ▶ Total angular momentum $\mathbf{L}$

Interaction Potentials – Pair Approximation

Pairwise additive interactions:

$$U(\mathbf{r}^N) = \sum_{i < j} u(r_{ij}), \quad r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$$

Forces:

$$\mathbf{F}_i = \sum_{j \neq i} f(r_{ij}) \hat{\mathbf{r}}_{ij}, \quad f(r) = -\frac{du}{dr}$$

Common pair potentials

- ▶ Hard sphere: $u = \infty$ if $r < d$
- ▶ Soft sphere: $u \propto r^{-n}$
- ▶ **Lennard-Jones** (today)
- ▶ Coulomb: $u \propto q_i q_j / r$
- ▶ Morse: diatomic bonds

Newton's 3rd law

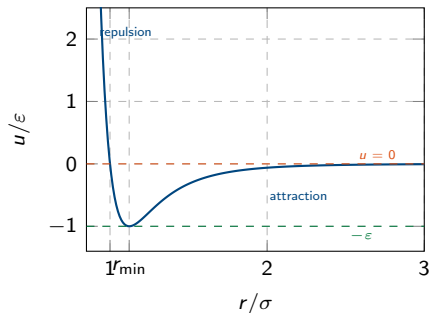
$$\mathbf{F}_{ij} = -\mathbf{F}_{ji}$$

Loop over $N(N-1)/2$ pairs
 $\Rightarrow$ halves computation.

For $N = 10^4$: $\sim 5 \times 10^7$ pairs.
 $\Rightarrow$ need **cutoff** & **neighbour lists**.

The Lennard-Jones Potential

$$u_{\text{LJ}}(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right]$$



Parameters

- ▶ ϵ – well depth (energy scale)
- ▶ σ – diameter (length scale)

Key features

- ▶ $u(r=\sigma) = 0$
- ▶ $r_{\min} = 2^{1/6}\sigma \approx 1.122\sigma$
- ▶ $u_{\min} = -\epsilon$
- ▶ r^{-12} : Pauli repulsion (empirical)
- ▶ r^{-6} : London dispersion

Argon parameters

$$\epsilon/k_B = 119.8 \text{ K}$$

$$\sigma = 3.405 \text{ \AA}, \quad m = 39.95 \text{ amu}$$

LJ Force and Reduced Units

Pair force along $\hat{\mathbf{r}}_{ij}$:

$$f(r) = -\frac{du_{\text{LJ}}}{dr} = \frac{48\varepsilon}{r} \left[\left(\frac{\sigma}{r}\right)^{12} - \frac{1}{2} \left(\frac{\sigma}{r}\right)^6 \right], \quad \mathbf{F}_{ij} = f(r_{ij}) \frac{\mathbf{r}_i - \mathbf{r}_j}{r_{ij}}$$

Reduced (LJ) units – set
 $m=\sigma=\varepsilon=1$

Quantity	LJ unit
Length	σ
Energy	ε
Mass	m
Time	$\tau = \sigma \sqrt{m/\varepsilon}$
Temperature	ε/k_B
Pressure	ε/σ^3

Why reduced units?

No unit conversions in code.
Results scale to *any* LJ fluid.

In reduced units:

$$u^* = 4[(r^*)^{-12} - (r^*)^{-6}]$$

Cutoff and Periodic Boundary Conditions

Potential cutoff at r_c (typically 2.5σ)

For continuity of potential energy:

$$\begin{cases} u(r) \rightarrow u(r) - u(r_c), & r < r_c \\ u(r) = 0, & r \geq r_c \end{cases}$$

Reduces computational cost.

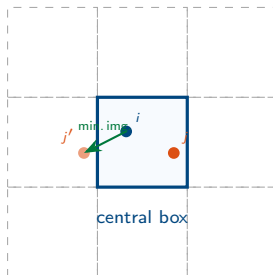
Periodic Boundary Conditions (PBC)

Minimum image convention:

$$\Delta r_\alpha \rightarrow \Delta r_\alpha - L \text{round}\left(\frac{\Delta r_\alpha}{L}\right)$$

Requirement: $r_c < L/2$.

Particles leaving one face re-enter from the opposite face.



Numerical Integration – Requirements

We need to solve:

$$\dot{\mathbf{r}}_i = \mathbf{v}_i, \quad \dot{\mathbf{v}}_i = \frac{\mathbf{F}_i}{m_i}$$

Desired properties of an MD integrator

- ① **Accuracy:** small local truncation error
- ② **Stability:** no energy drift over long runs
- ③ **Symplectic:** preserves phase-space volume
- ④ **Time-reversible:** mirrors physics
- ⑤ **Efficient:** few force evaluations per step

Why not Runge-Kutta?

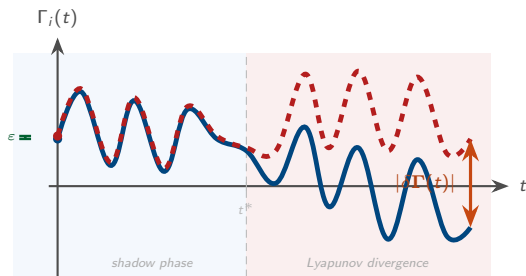
RK4 is accurate but **not symplectic**.

⇒ Systematic energy drift – unacceptable for MD.

Standard: Verlet family

2nd-order, time-reversible, symplectic integrators.
Conserve a *shadow Hamiltonian*.

Lyapunov Instability of MD Trajectories



Two trajectories, **nearby** I.C.:

- ▶ $\Gamma(0)$ — reference
- ▶ $\Gamma(0) + \delta\Gamma(0)$
perturbed, $\varepsilon = |\delta\Gamma(0)| \ll 1$

Initially *shadow* each other;
then **diverge exponentially**:

Lyapunov Divergence

$$|\delta\Gamma(t)| = \varepsilon e^{\lambda_1 t}$$

$\lambda_1 > 0$: **largest Lyapunov exponent**

$\Rightarrow$ Long-time trajectories not
reproducible; only *ensemble*
averages are meaningful.

Phase-Space Conservation – Liouville's Theorem

Liouville's theorem: Hamiltonian flow preserves phase-space volume.

$$\frac{d\Omega}{dt} = 0 \iff \nabla_{\Gamma} \cdot \dot{\Gamma} = 0$$

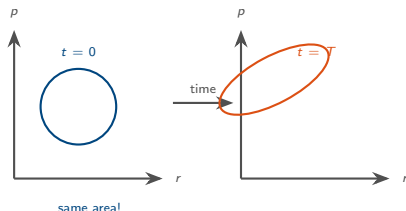
where Ω is the volume in phase space and Γ is the phase space coordinate $(\mathbf{r}, \mathbf{p})$ along the trajectories.

Symplectic integrators preserve volume/area in phase space

- ▶ No energy drift
- ▶ *Shadow Hamiltonian* conserved exactly

Non-symplectic (Euler, RK4):

- ▶ Violate Liouville's theorem
- ▶ $\Rightarrow$ exponential energy drift



Key point

Shape deforms; volume is conserved.

The Verlet Algorithm

Derivation – Taylor-expand positions forward and backward in time:

$$\mathbf{r}(t + \Delta t) = \mathbf{r}(t) + \mathbf{v}\Delta t + \frac{1}{2}\mathbf{a}\Delta t^2 + \frac{1}{6}\ddot{\mathbf{r}}\Delta t^3 + \mathcal{O}(\Delta t^4)$$

$$\mathbf{r}(t - \Delta t) = \mathbf{r}(t) - \mathbf{v}\Delta t + \frac{1}{2}\mathbf{a}\Delta t^2 - \frac{1}{6}\ddot{\mathbf{r}}\Delta t^3 + \mathcal{O}(\Delta t^4)$$

Adding the two equations:

$$\mathbf{r}(t + \Delta t) = 2\mathbf{r}(t) - \mathbf{r}(t - \Delta t) + \mathbf{a}(t)\Delta t^2 + \mathcal{O}(\Delta t^4)$$

$$\mathbf{v}(t) = \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t - \Delta t)}{2\Delta t} + \mathcal{O}(\Delta t^2)$$

Pros

- ▶ $\mathcal{O}(\Delta t^4)$ error in positions
- ▶ Time-reversible, symplectic
- ▶ One force evaluation per step

Cons

- ▶ Velocities not in integration step

The Velocity Verlet Algorithm

Velocity Verlet – four sub-steps

① **Half-kick:**

$$\mathbf{v}(t + \frac{\Delta t}{2}) = \mathbf{v}(t) + \frac{\Delta t}{2} \mathbf{a}(t)$$

② **Drift:**

$$\mathbf{r}(t + \Delta t) = \mathbf{r}(t) + \mathbf{v}(t + \frac{\Delta t}{2}) \Delta t$$

③ **New forces:**

$$\mathbf{a}(t + \Delta t) = \mathbf{F}(\mathbf{r}(t + \Delta t)) / m$$

④ **Half-kick:**

$$\mathbf{v}(t + \Delta t) = \mathbf{v}(t + \frac{\Delta t}{2}) + \frac{\Delta t}{2} \mathbf{a}(t + \Delta t)$$

Equivalent one-step form:

$$\mathbf{r}(t + \Delta t) = \mathbf{r} + \mathbf{v} \Delta t + \frac{1}{2} \mathbf{a} \Delta t^2$$

$$\mathbf{v}(t + \Delta t) = \mathbf{v} + \frac{1}{2} [\mathbf{a}(t) + \mathbf{a}(t + \Delta t)] \Delta t$$

This is the algorithm we implement.

Positions: $\mathcal{O}(\Delta t^4)$

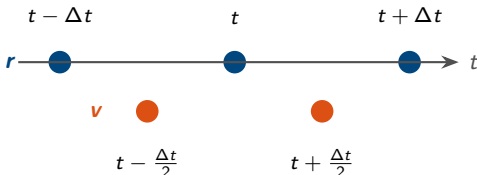
Velocities: $\mathcal{O}(\Delta t^2)$

Symplectic & reversible ✓

Leapfrog Algorithm

Algebraically equivalent to Verlet and velocity Verlet, staggered in time:

$$\mathbf{v}\left(t + \frac{\Delta t}{2}\right) = \mathbf{v}\left(t - \frac{\Delta t}{2}\right) + \mathbf{a}(t) \Delta t, \quad \mathbf{r}(t + \Delta t) = \mathbf{r}(t) + \mathbf{v}\left(t + \frac{\Delta t}{2}\right) \Delta t$$



Positions and velocities *leapfrog* past each other.

Verlet family

Verlet, velocity-Verlet and leapfrog lead to the same trajectories.

Velocity Verlet vs. Leapfrog

	VV	LF
Symplectic	✓	✓
r -error	$\mathcal{O}(\Delta t^4)$	$\mathcal{O}(\Delta t^4)$
Sync. velocities	✓	×

Choosing the Time Step

Rule of thumb: $\Delta t \ll \tau_{\text{vib}}$

For LJ fluids:

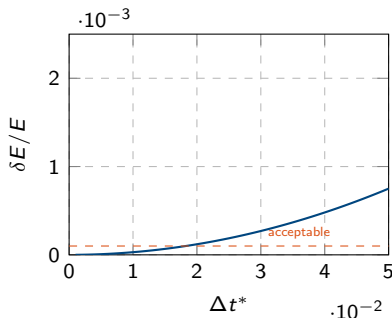
$$\Delta t^* = \Delta t / \tau \approx 0.001 - 0.005$$

Comparison with real systems:

$$\tau = \sigma \sqrt{m/\epsilon} \approx 2.2 \text{ ps (Ar)}$$

$$\Rightarrow \Delta t \approx 2 - 10 \text{ fs}$$

Diagnostic: $\delta E/E \lesssim 10^{-4}$ acceptable.



Large Δt : energy drift.
Small Δt : expensive.
Balance accuracy vs. cost.

Monitoring Energy Conservation

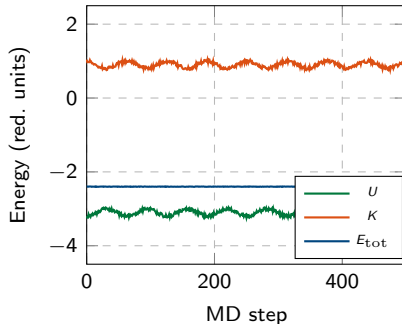
Total energy $E = K + U$ and momentum $\mathbf{P}_{\text{tot}} = \sum_i m_i \mathbf{v}_i$ must be conserved in NVE-MD.

Kinetic energy:

$$K = \frac{1}{2} \sum_{i=1}^N m_i v_i^2$$

Potential energy:

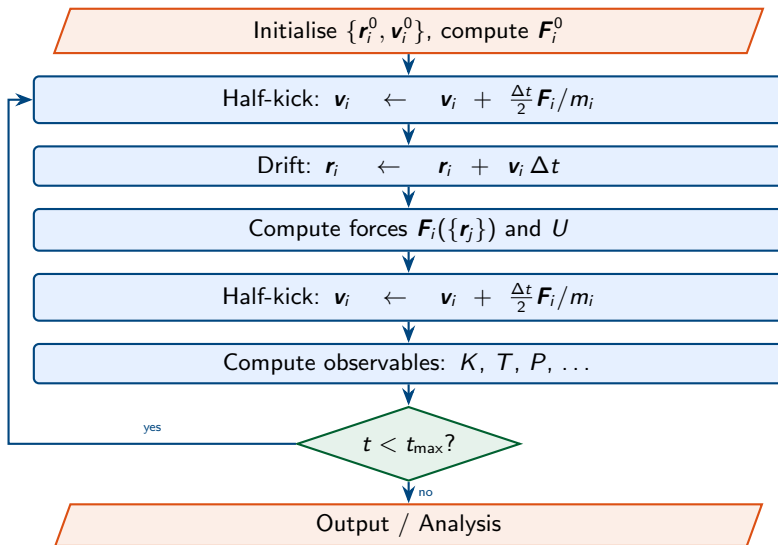
$$U = \sum_{i < j} u(r_{ij})$$



Diagnostic checklist

$K + U = \text{const} \checkmark$ $\mathbf{P}_{\text{tot}} = 0 \checkmark$ $T^* \approx T_{\text{target}}$ after equilibration $\checkmark$

The Basic MD Algorithm



Positions – Cubic lattice

- ▶ $N = 4n^3$ atoms, n unit cells per side
- ▶ Box length: $L = (N/\rho)^{1/3}$
- ▶ Ensures $r_{ij} > \sigma$ (no overlap)

Velocities – Maxwell-Boltzmann

$$v_{i,\alpha} \sim \mathcal{N}\left(0, \sqrt{k_B T/m}\right)$$

- 1 Subtract COM velocity:
 $\mathbf{v}_i \leftarrow \mathbf{v}_i - \bar{\mathbf{v}}$
- 2 Rescale to exact T

Always equilibrate first!

(Hundreds to thousands of steps before collecting statistics.)

Computing Thermodynamic Observables

Temperature

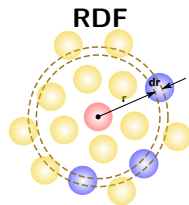
$$k_B T = \frac{2\langle K \rangle}{3N_f}, \quad N_f = 3N - 3$$

Pressure (virial theorem)

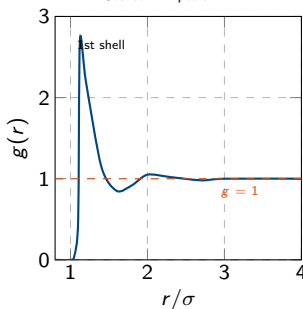
$$P = \frac{Nk_B T}{V} + \frac{1}{3V} \underbrace{\left\langle \sum_{i < j} \mathbf{r}_{ij} \cdot \mathbf{F}_{ij} \right\rangle}_{\text{Virial } \Xi}$$

Radial distribution function (RDF)

$$g(r) = \frac{\langle \rho(r) \rangle}{\rho} = \frac{1}{\rho} \left\langle \frac{1}{N} \sum_i \frac{1}{4\pi r^2} \sum_{j \neq i} \delta(r - r_{ij}) \right\rangle$$



Source: wikipedia



Mean-Square Displacement and Diffusion

Mean-square displacement (MSD):

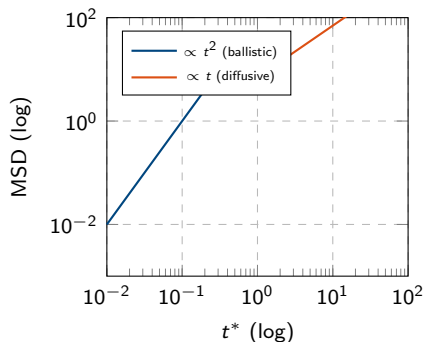
$$\text{MSD}(t) = \left\langle \frac{1}{N} \sum_{i=1}^N |\mathbf{r}_i(t) - \mathbf{r}_i(0)|^2 \right\rangle$$

Einstein relation:

$$D = \lim_{t \rightarrow \infty} \frac{\text{MSD}(t)}{6t}$$

Dynamical regimes:

- ▶ **Ballistic:** $\text{MSD} \propto t^2$
- ▶ **Diffusive:** $\text{MSD} \propto t$
- ▶ **Subdiffusive (glass):** t^α , $\alpha < 1$



Scaling and Optimisation

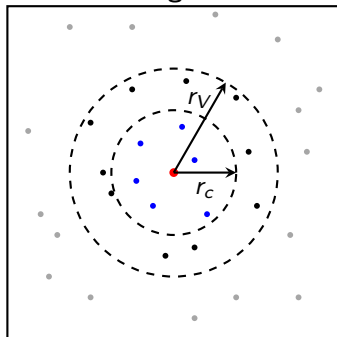
Naive force loop: $\mathcal{O}(N^2)$

- ▶ Feasible for $N \lesssim 10^4$
- ▶ $N = 10^3$: $\sim 5 \times 10^5$ pairs/step

Verlet neighbour list

- ▶ List pairs within $r_V = r_c + r_{\text{skin}}$
- ▶ Rebuild every ~ 20 steps
- ▶ $r_{\text{skin}} \approx 0.3\sigma$
- ▶ single step $\mathcal{O}(N)$, but rebuild

Verlet neighbour list



For production runs optimized software available
GROMACS, NAMD, OpenMM, CHARMM, LAMMPS, AMBER, etc.

Scaling and Optimisation

Naive force loop: $\mathcal{O}(N^2)$

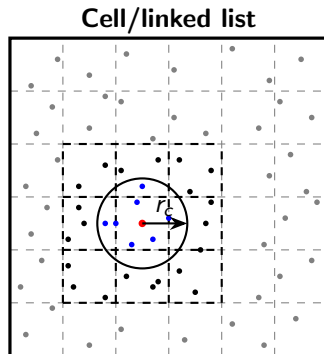
- ▶ Feasible for $N \lesssim 10^4$
- ▶ $N = 10^3$: $\sim 5 \times 10^5$ pairs/step

Verlet neighbour list

- ▶ List pairs within $r_V = r_c + r_{\text{skin}}$
- ▶ Rebuild every ~ 20 steps
- ▶ $r_{\text{skin}} \approx 0.3\sigma$
- ▶ single step $\mathcal{O}(N)$, but rebuild

Cell / linked-list method

- ▶ Cells of side $\geq r_c$
- ▶ Check 27 adjacent cells only (14 to avoid double counting)
- ▶ Strictly $\mathcal{O}(N)$



For production runs optimized software available
GROMACS, NAMD, OpenMM, CHARMM, LAMMPS, AMBER, etc.

Concepts introduced today

- ▶ MD = deterministic NVE simulation
- ▶ Ergodic hypothesis
- ▶ LJ potential: ε , σ , $r_{\min}$
- ▶ Reduced LJ units
- ▶ PBC, minimum image, cutoff
- ▶ Integrators
 - Requirements (Lyapunov instability, phase space measure)
 - Verlet family of algorithms
 - Symplectic, time-reversible
 - $\mathcal{O}(\Delta t^4)$ in positions
- ▶ Energy conservation as diagnostic
- ▶ Observables: T , P , $g(r)$, MSD
- ▶ Optimization strategies (Verlet and Cell lists)